

AP Computer Science Principles: The Complete Exam Guide

Exam format, the Create Task checklist, the 5 Big Ideas, a month-by-month plan, and 10 harder practice questions with full explanations.

Hi, I'm **Pigo!** I'll drop tips all through this guide.



3 hrs

end-of-course exam
120 + 60 minutes

70+4

multiple-choice
questions
+ written prompts

30%

of your score comes
from the Create task

62–63%

scored 3 or higher
in recent years

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01 THE EXAM AT A GLANCE

Two parts, one score

Your AP score (1–5) combines a multiple-choice exam with a programming project you build during the year. The project counts for nearly a third of your score, so treat it as seriously as the test.

70%

Section I · End-of-course multiple-choice exam

30%

Section II · Create performance task

Exam structure

Section	Weight & timing	What you do
I · Multiple choice	70% of score 120 minutes 70 questions	57 single-select questions · 5 single-select questions based on a reading passage about a computing innovation · 8 multiple-select questions where you choose exactly 2 answers. Every question has 4 options.
II · Create task: project	30% of score (with the written responses) 9 hours in class	Build a program of your choice. Submit three components: your program code , a video of it running, and your Personalized Project Reference (screenshots of key code segments).
II · Create task: written responses	60 minutes, on exam day	Answer 2 written-response questions (4 prompts in total) about the program you built, using your Personalized Project Reference.

FULLY DIGITAL

The exam is taken in Bluebook

The whole exam is taken in College Board's **Bluebook** app. An exam reference sheet explains the AP pseudocode used in questions, so you don't need to memorize the notation.

SUBMITTED BEFORE THE EXAM

Your project has a late-April deadline

Code, video and Personalized Project Reference are uploaded to the AP Digital Portfolio and must be marked **"submitted as final"** by the late-April deadline (11:59 PM ET). Check College Board for this year's exact date.

How to use this guide

Just starting?

Read pages 2–6, then follow the roadmap on page 8 month by month.

Mid-year?

Check the traps on page 7 and drill the free-response playbook on page 6.

Exam in weeks?

Take the 10 questions on page 9 in 20 minutes, then use the final-four-weeks plan.

SECTION I

02 The multiple-choice exam

70 questions in 120 minutes gives you about 1 minute 40 seconds per question. Most of the time pressure comes from code-tracing questions, so practice those under a timer.

57

single-select questions: one correct answer out of four

5

questions about a short reading passage on a computing innovation

8

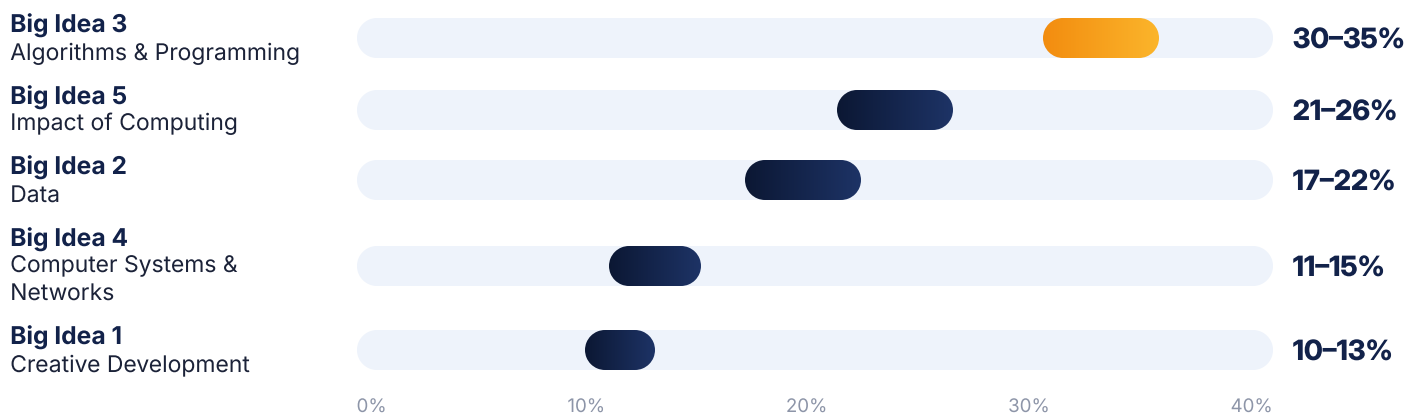
multiple-select: choose exactly 2; both must be right

0

points lost for a wrong answer, so never leave one blank

How the questions are weighted by Big Idea

College Board gives a range for each Big Idea's share of the multiple-choice score. Bars show minimum to maximum.



What this means for your study time: Algorithms & Programming is the biggest single area, but Impact of Computing is the second largest and the quickest to improve. It is mostly reading and reasoning, with no code to trace.

Strategies that work on this section

- Trace code on paper or the scratch area. Write each variable's value after every line, especially inside loops.
- AP lists start at index 1, not 0 like Python or Java. This is a common trap.
- REPEAT UNTIL checks first. If the condition is already true, the body never runs.
- For multiple-select questions, test each option on its own as true or false before choosing your two.
- Reading-passage questions reward careful reading over memorized facts. Find the line that proves your answer.
- Don't leave anything blank. There is no penalty for wrong answers.

03 WHAT THE COURSE COVERS

The 5 Big Ideas

Everything on the exam fits into one of these five areas.

BIG IDEA

1

10–13%

Creative Development

How programs are designed, built, tested and improved, often by teams.

Be ready to: describe a program's purpose, inputs and outputs; explain how collaboration helps; find and fix syntax, logic and run-time errors; choose good test cases.

BIG IDEA

2

17–22%

Data

How computers store information in binary and turn data into useful information.

Be ready to: convert binary and decimal; explain overflow and rounding errors; compare lossless and lossy compression; explain metadata; describe cleaning, filtering and finding patterns in data.

BIG IDEA

3

30–35%

Algorithms & Programming

Writing and reading code: variables, conditions, loops, lists and procedures.

Be ready to: trace selection and iteration; work with lists; use procedures with parameters; compare algorithm efficiency; explain binary search, simulations and heuristics; recognize undecidable problems.

BIG IDEA

4

11–15%

Computer Systems & Networks

How the internet moves data, and how work is split across computers.

Be ready to: explain packets, protocols and routing; explain why redundancy makes the internet fault tolerant; calculate sequential, parallel and distributed run times and speedup.

BIG IDEA

5

21–26%

Impact of Computing

The effects of computing on people and society, good and bad.

Be ready to: discuss the digital divide and bias; explain crowdsourcing, licensing and intellectual property; recognize phishing, malware, keylogging and the role of encryption and multifactor authentication.

◆ The skills College Board assesses

1 • Computational solution design

Design and evaluate solutions to a problem.

2 • Algorithms & program development

Develop and implement algorithms.

3 • Abstraction in program development

Use procedures and lists to manage complexity.

4 • Code analysis

Evaluate and test algorithms and programs.

5 • Computing innovations

Investigate the effects of innovations.

6 • Responsible computing

Taught in the course but not directly scored on the exam.

SECTION II

04

The Create performance task

You get 9 hours of class time to build a program of your choice. Use these checklists before you submit. Each item is a College Board requirement.

1

Program code must include

Six requirements

Input (user, device, data stream or file) · at least one **list** that manages complexity · a **procedure you wrote** with one or more **parameters** · an **algorithm** in that procedure using sequencing, selection *and* iteration · **calls** to the procedure · **output** based on input.

Test: would the program be harder to write without the list? If not, it isn't managing complexity.

2

Video must

Check before uploading

Be **1 minute or less** and **30 MB or less** · use .webm, .mp4, .wmv, .avi or .mov · show input, some functionality and output · have **no voice narration** (captions are fine) · show **nothing that identifies you**.

Record with captions only, then check the file size before submitting.

3

Personalized Project Reference

Your only notes on exam day

Two code segments for your **procedure** (its definition and a call) and two for your **list** (where data is stored and where it is used). Clear screenshots, text at least **10-point**, and **no comments**.

College Board warns that comments or extra text in the PPR can score 0 for the whole task.

4

Academic integrity

What help is allowed

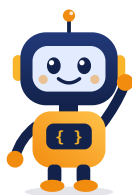
You may collaborate with a partner while *developing* the program, but the video, PPR and written responses must be your own. Teachers and tutors can teach concepts and help on *other* projects, not on the work you submit.

Project ideas: quiz game, study-time tracker, playlist filter. Make the code and design your own.

Plan your 9 in-class hours

Plan 1 hr	Build & test 5 hrs	Video 1 hr	PPR 1 hr	Final check 1 hr
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A suggested split. Your teacher decides when the 9 in-class hours happen, so start planning early.



Mark **all three** components "submitted as final" well before the late-April deadline. Uploading alone isn't enough!

SECTION II ON EXAM DAY

05 The written responses

On exam day you get 60 minutes for 2 questions (4 prompts) about your own program. Here is a strong sample answer to an algorithm prompt.

PRACTICE TASK • WRITTEN BY ACT SAT GO

Practice prompt (Written Response 2a style): Consider the procedure in your Personalized Project Reference. Describe in detail how the algorithm in the procedure works, and explain how the result would change if the selection statement were removed. Sample program: a study tracker with procedure `checkGoal(subject)` and list `minutesLog`.

Sample response

- ① My procedure `checkGoal(subject)` takes a subject name as its parameter. It sets `total` to 0, then uses a **FOR EACH** loop to visit every entry in the list `minutesLog`.
- ② Inside the loop, an **IF** statement checks whether the entry's subject matches the parameter. Only matching entries are added to `total`.
- ③ After the loop, a second IF compares `total` to the goal of 120 minutes and returns "on track" or "behind".
- ④ Without the first IF, every entry would be added, so `total` would be the minutes for *all* subjects and the procedure would say "on track" too often.

- ① **Name your code.** Refer to the procedure and list by their real names.
- ② **Describe the iteration:** what the loop visits and what happens on each pass.
- ③ **Describe the selection:** what is checked and what it changes.
- ④ **Answer the "what if"** with the specific new output, not a general definition.

◆ Five habits for the written responses

- 1** Answer every part of the prompt in order.
- 2** Talk about *your* program, not definitions.
- 3** Use the exact names from your PPR.
- 4** Give specific inputs and outputs for testing prompts.
- 5** Practice each prompt type in 8 minutes or less.

06 LEARN FROM OTHERS

10 traps that cost points

Meet **Glitch**. He loves these ten mistakes. Squash each one before May.

Index 0? **Not in AP pseudocode!**



01 Comments in the PPR

Comments or extra text in the Personalized Project Reference can score 0 on the Create task. Screenshot clean code only.

02 A parameter that does nothing

The parameter must affect what the procedure does. An unused parameter doesn't meet the requirement.

03 Iteration outside the procedure

Sequencing, selection *and* iteration must all be inside the procedure you write about.

04 A list that doesn't manage complexity

If a few separate variables would work just as well, your list isn't doing its job.

05 A video that breaks the rules

Over 1 minute, voice narration, or your name or face on screen. Check it against the checklist.

06 Forgetting "submit as final"

Uploading isn't enough. All three components must be marked final before the deadline.

07 Starting list indexes at 0

In AP pseudocode the first item is `list[1]`. Python and Java habits cost points.

08 Misreading REPEAT UNTIL

The condition is checked *before* each pass, and the loop stops as soon as it is true.

09 One answer on multiple-select

Eight questions need exactly two answers, and you need both right to earn the point.

10 Underestimating Big Idea 5

Impact of Computing is about a quarter of the multiple-choice section. Review bias, the digital divide, licensing, privacy and security.

Quick self-check: can you explain every line of your Personalized Project Reference to a friend without looking at the rest of your program? If not, that's the first thing to practice.

07

PLAN YOUR YEAR

Your study roadmap to May

A steady plan beats a last-minute rush. Adjust the months to your teacher's schedule. Your teacher decides when the 9 in-class Create hours happen.



Sep – Dec
Foundations



Big Ideas 1 & 3 **biggest area**

Learn variables, conditions, loops, lists and procedures. Trace 5 short pseudocode snippets a week, and get used to lists starting at index 1.

Jan – Feb
Widen & rehearse



Big Ideas 2, 4 & 5 + two practice programs

Binary, compression, networks and the impact of computing. Build two small programs that meet all six code requirements as a rehearsal.

Mar – Apr
Create task



Build, record, screenshot, submit **late-April deadline**

Complete your program in class. Record the video, build a clean PPR, and mark everything final well before the deadline.

Late April
Practice



Timed multiple-choice sets

Take a full 70-question practice set in 120 minutes. Review every miss by Big Idea.

Early May
Final review



Written-response drills

Write timed answers to practice prompts using your own PPR, 8 minutes each.

Exam day
May



AP CSP exam · Bluebook

Sleep well, charge your device, and use the checklist at the end of this guide.

◆ The final four weeks

Week	Focus	Target
4 weeks out	Diagnose: one timed set of 35 questions, then review every mistake	List your three weakest Big Ideas
3 weeks out	Fix your weakest area; make sure the Create task is submitted as final	Create task done; 80%+ on drills
2 weeks out	Full 70-question timed practice; written-response drills	Finish within 120 minutes
Final week	Light review: Big Idea 5 vocabulary, list indexing, loop tracing	Rest the night before

08 TEST YOURSELF 10 practice questions

Original questions written by ACT SAT GO in the style of the exam, pitched at the harder end. Give yourself **20 minutes** for all ten, then check page 11.

1 BIG IDEA 3 · LISTS & PROCEDURES

```
PROCEDURE mystery(list)
{
  result ← []
  FOR EACH item IN list
  {
    IF (item MOD 3 = 0)
    {
      INSERT(result, 1, item)
    }
    ELSE
    {
      APPEND(result, item)
    }
  }
  RETURN(result)
}
```

What is displayed by `DISPLAY(mystery([2, 6, 7, 9, 12]))` ?

- A [6, 9, 12, 2, 7]
- B [2, 7, 6, 9, 12]
- C [12, 9, 6, 7, 2]
- D [12, 9, 6, 2, 7]

2 BIG IDEA 3 · LOOPS

What is displayed when this code segment runs?

```
x ← 10
y ← 3
REPEAT UNTIL (x < y)
{
  x ← x - y
  y ← y + 1
}
DISPLAY(x)
DISPLAY(y)
```

- A 7 4
- B -2 6
- C 3 5
- D 3 4

3 BIG IDEA 3 · BOOLEAN LOGIC

Which expression is `true` if and only if **exactly one** of the Boolean variables `p` and `q` is `true` ?

- A $(p \text{ OR } q) \text{ AND NOT } (p \text{ AND } q)$
- B $(p \text{ AND } q) \text{ OR NOT } (p \text{ OR } q)$
- C $\text{NOT } p \text{ OR NOT } q$
- D $(p \text{ OR } q) \text{ AND } (p \text{ AND } q)$

08 TEST YOURSELF (continued)

4 BIG IDEA 2 · OVERFLOW

A system stores whole numbers using 4 bits with no sign bit. What happens when it adds the binary numbers

1011 and 0110?

- A The result is 10001, stored correctly
- B A round-off error occurs because binary cannot store 17
- C The result is 1101, which equals 13
- D The sum, 17, needs more than 4 bits, so an overflow error occurs

5 BIG IDEA 3 · EFFICIENCY

A **sorted** list contains 500 numbers. What is the greatest number of list elements a binary search would examine to find a value or determine it is absent?

- A 8
- B 9
- C 250
- D 500

6 BIG IDEA 4 · PARALLEL COMPUTING

A program has four steps. P takes 40 s, Q takes 30 s and R takes 50 s; these three are independent. Step S takes 20 s and can start only after **both** P and Q finish.

What is the minimum time to run the program on **two** processors, and what is the speedup compared with one processor?

- A 70 s; speedup 2.0
- B 80 s; speedup 1.75
- C 90 s; speedup about 1.56
- D 140 s; speedup 1.0

7 BIG IDEA 4 · FAULT TOLERANCE

A network has these direct connections: A–B, A–C, B–D, C–D, D–F, C–E, E–F. What is the **minimum** number of connections that must fail for A and F to be unable to communicate?

- A 1
- B 2
- C 3
- D 4

8 BIG IDEA 2 · COMPRESSION

A team must send scanned legal contracts so that every pixel can be restored **exactly**, while using as little storage as possible. Which choice is best?

- A Lossy compression, because it always gives a smaller file
- B No compression, because compressed files cannot be restored
- C Lossless compression, because exact reconstruction is required
- D Lossy compression, because it removes metadata only

9 BIG IDEA 5 · BIAS IN DATA

A city trains a model to decide where to send repair crews, using complaints filed through a smartphone app. Which change would **most directly** reduce bias in the results?

- A Add phone-in and in-person reporting so residents without smartphones are counted
- B Use a faster server to process complaints
- C Encrypt the complaint data
- D Collect more complaints from the neighborhoods that already report most

10 BIG IDEA 3 · HEURISTICS

A delivery app must choose the shortest route through 40 stops. Checking every possible ordering would take far too long. Why might the developers use a heuristic?

- A It always finds the shortest route faster
- B The problem is undecidable, so no algorithm can produce any route
- C It finds a good-enough route in reasonable time, without guaranteeing the best one
- D It turns an unreasonable-time problem into a sorting problem

09

CHECK YOUR WORK

Answer key & explanations

Read every explanation, even for questions you got right. Each one points to a pattern that keeps coming back on the exam.

#	Answer	Why
1	D · [12, 9, 6, 2, 7]	Multiples of 3 are inserted at the front, so they end up in reverse order: 6, then 9 in front of it, then 12 in front of that. 2 and 7 are appended in order: [12, 9, 6, 2, 7].
2	C · 3 5	x, y go (10, 3) → (7, 4) → (3, 5). The condition is checked before each pass; at (3, 5), $x < y$ is true, so the loop stops and 3 and 5 are displayed.
3	A	This is "exclusive or": at least one is true, but not both. Choice C is true when both are false, and B is true when both are equal.
4	D	1011 is 11 and 0110 is 6, so the sum is 17. The largest 4-bit value is 15, so the result cannot be represented: an overflow error. Round-off errors involve real numbers, not whole numbers.
5	B · 9	Binary search halves the remaining list each step. $2^8 = 256 < 500 \leq 512 = 2^9$, so at most 9 elements are examined.
6	B · 80 s; 1.75	Best plan: processor 1 runs Q (0–30) then R (30–80); processor 2 runs P (0–40) then S (40–60). Total 80 s. One processor takes $40 + 30 + 50 + 20 = 140$ s, so speedup = $140 \div 80 = 1.75$. 70 s is impossible because S must wait.
7	B · 2	Several routes join A to F (e.g., A-B-D-F, A-C-D-F, A-C-E-F). Every route ends with D-F or E-F, so removing those two cuts F off. No single failure blocks every route.
8	C	Only lossless compression guarantees the original can be rebuilt exactly. Lossy compression gives smaller files by permanently discarding information.
9	A	The data under-represents residents without smartphones (the digital divide). Adding other reporting methods fixes who is counted. Choice D makes the bias worse.
10	C	A heuristic trades a guarantee of the best answer for an answer found in reasonable time. The route problem is decidable; it just takes unreasonable time to solve exactly.

9–10

Strong. Move on to full timed free-response practice.

6–8

On track. Revisit the unit behind every miss on page 4.

0–5

Build foundations now. Start with tracing loops and lists (Big Idea 3).

10 KNOW THE TARGET How students score

In recent years, about 62–63% of students earned a 3 or higher on AP Computer Science Principles, and about 10–11% earned a 5. A strong Create task and careful multiple-choice work are what separate a 4 from a 5.

62–63%

of students earned a 3 or higher in recent years

10–11%

earned the top score of 5 in recent years

3+

is the score many colleges accept for credit or placement

What each AP score means

5

Extremely well qualified

4

Very well qualified

3

Qualified

2

Possibly qualified

1

No recommendation

Ranges reflect recent College Board score distributions and shift a little each year. Credit and placement policies vary by college, so check each college's AP policy.

Exam-day checklist

- ☐ **Bluebook** is installed and updated, and your device passed the exam check
- ☐ Device **fully charged**, with the charger packed
- ☐ Admission ticket and photo ID, as your school requires
- ☐ Confirm your **exam day and start time** in May with your AP coordinator
- ☐ Your Create task was **submitted as final** before the late-April deadline
- ☐ Re-read your **Personalized Project Reference** the night before
- ☐ Plan your time: about **1 min 40 s** per multiple-choice question; about **15 min** per written prompt
- ☐ On multiple-select questions, choose **exactly two** answers

In the written responses: refer to the procedure and list names from your own code, describe what *your* program does rather than giving general definitions, and answer each part of the prompt in order.



ACT SAT GO • 1-ON-1 AP TUTORING

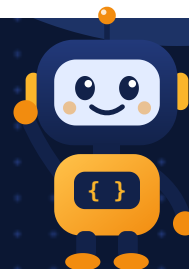
Aiming for a 5? Get a personal AP CSP plan.

Book a free diagnostic lesson. We'll find your weak Big Ideas, review your Create task timeline, and give you a week-by-week plan to exam day in May.

Book a free diagnostic lesson →



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Concept tutoring

Live 1-on-1 sessions on all five Big Ideas, with pseudocode tracing drills after every lesson.

Create task readiness

Practice projects and written-response drills. We teach the skills; your submitted project stays 100% your own work.

Exam practice

Full-length multiple-choice sets with score reports, so you see progress before May.

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